Balanced diet: learning reference

Useful sources are examples: foods can supply several components. Balance means suitable amounts for needs over time, not equal amounts.

Component	Useful function	Useful source
Carbohydrate	Energy for respiration	Rice, bread, potatoes
Protein	Growth and repair materials	Beans, lentils, eggs, fish
Fat / lipid	Energy store; insulation; vitamin absorption	Oil, nuts, seeds
Vitamins	Specific processes in small amounts	C: peppers; A precursor: carrots
Minerals	Specific processes, e.g. iron for haemoglobin	Iron: lentils; calcium: dairy/fortified alternatives
Fibre	Helps food move through the gut	Wholegrains, fruit, vegetables
Water	Medium for many reactions; transport; cooling	Drinking water; water in foods

Four required deficiency links

Nutrient	Body process and possible effect	Useful source
Iron	Haemoglobin helps transport oxygen; Deficiency can cause iron-deficiency anaemia	Lentils, beans, meat
Vitamin A	Needed for normal vision; Low intake can impair night vision; severe prolonged deficiency can cause blindness	Eggs; carrots supply beta-carotene, converted to vitamin A
Vitamin C	Healthy connective tissues and wound healing; Severe deficiency can cause scurvy	Peppers, citrus fruit
Vitamin D	Helps regulate calcium/phosphate for healthy bones; Deficiency can cause rickets in children	Oily fish, egg yolk, appropriately fortified foods

Starvation: prolonged inadequate energy intake. **Malnutrition:** deficiencies, excesses or imbalances of energy and/or nutrients. Starvation is one form.
Limits: not all anaemia is caused by iron deficiency. Rickets can also involve calcium deficiency. These are fictional evidence exercises, not medical diagnoses. Symptoms or a food photograph alone do not establish a deficiency.

Water is a medium for many reactions. The body can make vitamin D with suitable UV exposure; sunlight is not a food source. This sheet gives no personal diet, supplement or exposure prescription. Food-group proportions are different from these seven dietary components.

Practise, keep, then improve

Use the reference while learning. Complete the missing function/source links; protein is done. Keep original answers. Correct only wrong links in another pen.

Component	Useful function	Useful source
Carbohydrate	_____	_____
Protein	Growth and repair materials	Beans, lentils, eggs, fish
Fat / lipid	_____	_____
Vitamins	_____	_____
Minerals	_____	_____
Fibre	_____	_____
Water	_____	_____

Four possible effects

Low iron → _____; low A → _____; low C → _____; low D in a child → _____.

Guided cases: follow the supplied facts

Both records meet energy needs. Harbour: iron below needs for months; lentils provide iron. Garden: C below needs for months; peppers provide C.

Case	Nutrient gap	Possible effect	Justified change
Harbour	Iron below needs	_____	_____ because _____
Garden	_____	_____	_____ because _____

Original four-mark question

For months a fictional diet meets energy needs. Vitamin A stays below needs; other measured nutrients are adequate. Given: carrots supply beta-carotene, converted to vitamin A. Explain why this is malnutrition rather than starvation. State a possible effect and justify one change. [4]

Revision: code __; the scientific relationship I changed is:

Help: N nutrient role · L deficiency link · B balance evidence · J justified change. Watch the coded example before self-coding. Use the support cards, close content prompts for a different recheck, and turn this sheet away for protected fresh evidence. Keep usual access.